Shor's algorithm, connected

Order finding is phase estimation of modular multiplication. One state connects values, combs, eigenphases, and measurable frequencies.

1. Put the period in a spectrum. Handle even N and prime powers classically. For remaining composite N, choose $1 < a < N$. A nontrivial $\gcd(a, N)$ already gives a factor; otherwise seek $r = \min\{d > 0 : a^d \equiv 1 \pmod N\}$. Set $f(x) = a^x \bmod N$, $n = \lceil \log_2 N \rceil$, and $Q = 2^t$ with $N^2 \le Q < 2N^2$. Register C has t qubits, W has n; $0 \le x, y < Q$, $0 \le c, s < r$. The permutation $U_a|w\rangle = |aw \bmod N\rangle$, extended by identity on unused labels, cycles $|e_c\rangle = |a^c \bmod N\rangle$ with length r. For $\omega = e^{2\pi \mathrm{i}/r}$,

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{c=0}^{r-1} \omega^{-sc} |e_c\rangle, \qquad U_a|u_s\rangle = \omega^s |u_s\rangle, \qquad |1\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle.$$

An index shift and roots-of-unity orthogonality prove these identities. This unknown r-point Fourier basis is analytical; the circuit uses no unknown transform.

2. Make an eigenphase relative, efficiently. Control makes an unobservable global phase relative, sending $(\alpha|0\rangle + \beta|1\rangle)|u_s\rangle$ to $(\alpha|0\rangle + \beta\omega^s|1\rangle)|u_s\rangle$. An eigenstate target stays factored. For $x = \sum_j 2^j x_j$, the ladder $\Lambda = \sum_x |x\rangle\langle x| \otimes U_a^x$ uses t controlled multipliers by $b_j = a^{2^j} \bmod N$, computed by repeated squaring. Control phases $2^j s/r$ combine to ω^{sx}, rather than independently readable bits.

3. Use the easy input to unite three representations. From $|0\rangle_C|1\rangle_W$, $H^{\otimes t}$ gives $Q^{-1/2}\sum_x |x\rangle|1\rangle$; then apply Λ. Define $M_c = \#\{x : x \equiv c \pmod r\} = 1 + \lfloor (Q - 1 - c)/r \rfloor$ and $|C_c\rangle = M_c^{-1/2} \sum_{m=0}^{M_c - 1} |c + mr\rangle$:

$$|\Psi_2\rangle = \underbrace{\frac{1}{\sqrt{Q}} \sum_x |x\rangle|e_{x \bmod r}\rangle}_{\text{input/output values}} = \underbrace{\sum_c \sqrt{M_c/Q}\,|C_c\rangle|e_c\rangle}_{\text{orthonormal Schmidt combs}}$$

$$= \underbrace{\frac{1}{\sqrt{r}} \sum_s |\Phi_s\rangle|u_s\rangle}_{\text{eigenphase ramps with orthogonal work tags}}, \qquad |\Phi_s\rangle = \frac{1}{\sqrt{Q}} \sum_x e^{2\pi \mathrm{i} s x/r} |x\rangle.$$

Values and eigenphases describe one permutation, rather than successive gates. Easy $|1\rangle$ avoids unknown eigenvector preparation. For $r > 1$, the joint state is entangled.

4. Entanglement determines the local coherences and information. Ignoring W gives

$$\rho_C = \mathrm{Tr}_W |\Psi_2\rangle\langle\Psi_2| = \sum_c \frac{M_c}{Q} |C_c\rangle\langle C_c| = \frac{1}{r} \sum_s |\Phi_s\rangle\langle\Phi_s|, \qquad (\rho_C)_{xx'} = \frac{1}{Q} \mathbf{1}_{x \equiv x' \,(\mathrm{mod}\, r)}.$$

The diagonal is flat: measuring now gives uniform x. Identical work records preserve interference within a residue class; orthogonal records remove local cross terms between classes. The Schmidt rank is r, the eigenvalues are $\lambda_c = M_c/Q$, and $E = S(C) = S(W) = -\sum_c \lambda_c \log_2 \lambda_c \le \log_2 r$ ebits, $S(CW) = 0$, $I(C:W) = S(C) + S(W) - S(CW) = 2E$, $S(C|W) = -E$. If $r \mid Q$, all $M_c = Q/r$ and $E = \log_2 r$: k Bell pairs have 2^k equal branches. When $r \nmid Q$, the ramp ensemble is not spectral: its weights $1/r$ are not Schmidt eigenvalues. Entropy measures information in the weights, not branch count or accessible classical bits.

5. Phase estimation reads the frequency by interference. Use the known Fourier matrix $(F_Q)_{xy} = Q^{-1/2} e^{2\pi \mathrm{i} xy/Q}$. Apply $F_Q^\dagger$: each y tests frequency y/Q, unwinding matching phases so amplitudes add. The QPE unitary is $(F_Q^\dagger \otimes \mathbb{I})\Lambda(H^{\otimes t} \otimes \mathbb{I})$, and

$$|\Psi_3\rangle = \frac{1}{\sqrt{r}} \sum_{y,s} D_Q(s/r - y/Q)|y\rangle|u_s\rangle, \qquad D_Q(\delta) = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi \mathrm{i} x\delta},$$

$$P(y) = \langle y|F_Q^\dagger \rho_C F_Q|y\rangle = \frac{1}{r} \sum_s |D_Q(s/r - y/Q)|^2, \quad |D_Q(\delta)|^2 = \frac{\sin^2(\pi Q\delta)}{Q^2 \sin^2(\pi\delta)}.$$

Use continuous limits at integer δ. Orthogonal u_s tags make counting statistics an exact uniform mixture of phase-estimation laws, while x-alternatives interfere within each law. This is a statistical equivalence, not hidden branch selection. If $r \mid Q$, outputs are $y = sQ/r$, each with probability $1/r$; otherwise peaks broaden. The inverse QFT need not diagonalize ρ_C and preserves E. Conditioned on y, the work state is $|w_y\rangle \propto \sum_s D_Q(s/r - y/Q)|u_s\rangle$, generally a superposition.

6. Reconstruct, verify, and factor. For a nearest-grid output, $|y/Q - s/r| \le 1/(2Q) < 1/(2r^2)$, so the reduced fraction appears among continued-fraction convergents. Such an output has probability at least $4/\pi^2$ per phase branch. Its denominator is $r/\gcd(s, r)$; coprime s recovers r. Repeat and verify candidates: $a^d \equiv 1 \pmod N$ generally proves $r \mid d$, not minimality alone. For even r and $z = a^{r/2} \not\equiv \pm 1 \pmod N$, $N \mid (z - 1)(z + 1)$ and $\gcd(z \pm 1, N)$ give factors. Retry samples or bases when necessary.

7. Preserve the right records. Extra scratch multiplies $(\rho_C)_{xx'}$ by $\langle g_{x'}|g_x\rangle$. Uncompute records distinguishing inputs inside one residue class, retaining $f(x)$. Reversing Λ restores the uniform input and forces $y = 0$. Measuring W's modular value and forgetting it removes $C|W$ entanglement but preserves ρ_C, comb coherence, and Fourier statistics. Arithmetic and reconstruction cost polynomially in $\log N$; the QFT uses $O(t^2)$ gates.

Detailed derivations: explainer Sections 3–8 and 11–15. Presentation connections and its four-branch example are integrated into the full text. All identities above allow $r \nmid Q$, except the explicitly stated exact-peak specialization.